

Chapter 1

Limits and Their Properties

(極限與其性質)

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Useful Notations

(常用的數學符號)



1. A set (集合) is a collection of specified objects, and usually denoted by $A, B, C, \dots$.

$$\mathbb{R} = \{x \mid x \text{ is a real number (實數)}\}$$

$$\mathbb{N} = \{x \mid x \text{ is a positive integer (正整數)}\}$$

$$\mathbb{Z} = \{x \mid x \text{ is an integer (整數)}\}$$

$$\mathbb{Q} = \{x \mid x \text{ is a rational number (有理數)}\}$$

2. $x \in A$: x belongs to (屬於) A , i.e., x is an element of the set A .

$$\pi \in \mathbb{R}, \quad 5 \in \mathbb{N}, \quad -7 \in \mathbb{Z} \quad \text{and} \quad \frac{2}{3} \in \mathbb{Q}.$$

$A \subseteq B$: A is a subset (子集合) of B , i.e., if $x \in A$, then $x \in B$.



3. The union (聯集) and intersection (交集) of two sets A and B are defined by

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\},$$

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

4. $\forall$: for all (對於所有).

5. $\exists$: exist (存在).

6. $\nexists$: does not exist (不存在).

7. s.t. or $\ni$: such that (使得).



8. $\implies$: imply that (意指).

9. $\iff$: if and only if (若且唯若).

10. Greek letters: (常用希臘字符)

α (alpha), β (beta), γ (gamma), δ (delta),
 ε (epsilon), θ (theta), λ (lambda), μ (mu),
 ρ (rho), τ (tau), ϕ (phi), ω (omega), $\dots$



11. The intervals (區間) in $\mathbb{R}$ are often denoted by I , e.g.,

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\},$$

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\},$$

$$(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\},$$

$$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}.$$



- 12. i.e.: that is (也就是說).
- 13. e.g.: for example (舉例來說).
- 14. w.r.t.: with respect to (關於).
- 15. Def: Definition (定義), Thm: Theorem (定理).



Section 1.2

Functions and Their Graphs

(函數與其圖形)



Def (實值函數的定義)

Let $X \subseteq \mathbb{R}$ and $Y \subseteq \mathbb{R}$.

- (1) A real-valued function f from X to Y , denoted by $f: X \rightarrow Y$, is a correspondence (對應) that assigns to (指派) each $x \in X$ one unique (唯一的) value $y \in Y$.
- (2) The set $X = \text{dom}(f)$ is called the domain (定義域) of f .
- (3) The subset of Y defined by

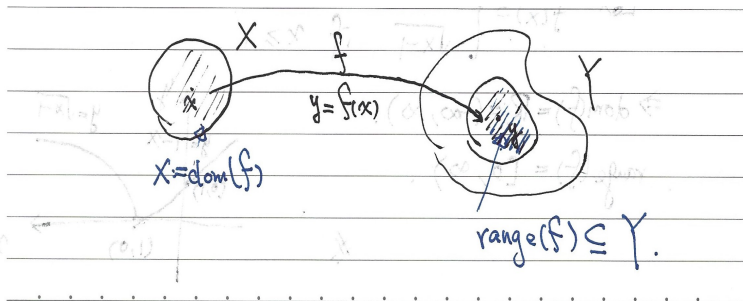
$$\text{range}(f) = \{y \in Y \mid \exists x \in X \text{ s.t. } y = f(x)\}$$

is called the range (值域) of f .

- (4) x is the independent variable (自變數) and y is the dependent variable (應變數) of f .



函數映射的示意圖

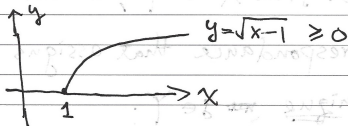


Example 2: Find $\text{dom}(f)$ and $\text{range}(f)$.

(a) $f(x) = \sqrt{x-1}$

$$\Rightarrow \text{dom}(f) = \{x \in \mathbb{R} \mid x-1 \geq 0\} = [1, \infty)$$

$$\text{range}(f) = \{f(x) \mid x \geq 1\} = [0, \infty)$$



(b) $f(x) = \tan x$

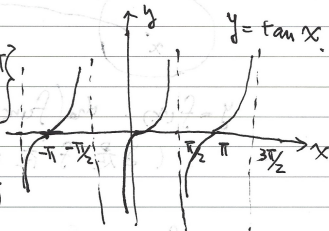
$$\Rightarrow \text{dom}(f) = \left\{ x \in \mathbb{R} \mid x \neq \left(n + \frac{1}{2}\right)\pi \right\},$$

with $n \in \mathbb{Z}$.

$$\text{range}(f) = \left\{ f(x) \mid x \neq \left(n + \frac{1}{2}\right)\pi \right\}$$

with $n \in \mathbb{Z}$

$$= \mathbb{R} = (-\infty, \infty)$$

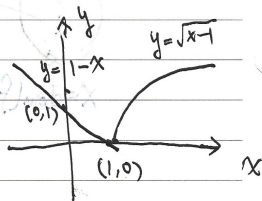


Example 3: (分段函数的定义域与值域)

$$\text{Let } f(x) = \begin{cases} 1-x & \text{if } x < 1 \\ \sqrt{x-1} & \text{if } x \geq 1 \end{cases}$$

$$\Rightarrow \text{dom}(f) = \mathbb{R} = (-\infty, \infty)$$

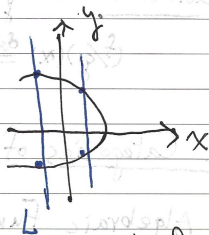
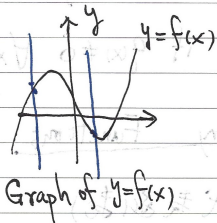
$$\text{range}(f) = [0, \infty)$$



* Graph of a Function

Thm: (Vertical Line Test)

Any vertical line intersects the graph of $y=f(x)$ at most exactly once.



Not ~~a~~ graph of a
function graph!



Def (函數的基本運算; 1/2)

Let f and g be real-valued functions defined on $X \subseteq \mathbb{R}$.

(1) The sum $f + g$ of f and g is defined by

$$(f + g)(x) = f(x) + g(x) \quad \forall x \in X.$$

(2) The difference $f - g$ of f and g is defined by

$$(f - g)(x) = f(x) - g(x) \quad \forall x \in X.$$

(3) For any $k \in \mathbb{R}$, the constant multiple kf of f is defined by

$$(kf)(x) = k \cdot f(x) \quad \forall x \in X.$$



Def (函數的基本運算; 2/2)

(4) The product fg of f and g is defined by

$$(fg)(x) = f(x) \cdot g(x) \quad \forall x \in X.$$

(5) The quotient $\frac{f}{g}$ of f and g is defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad \forall x \in X,$$

provided that $g(x) \neq 0 \quad \forall x \in X$.



Three Categories of Elementary Functions (1/2)

1. Algebraic Functions (代數函數)

(a) polynomial (function) of n th degree ($n \in \mathbb{N}$)

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \quad \text{with } a_n \neq 0.$$

(b) rational function (有理函數)

$$f(x) = p(x)/q(x),$$

where $p(x)$ and $q(x) \neq 0$ are polynomials.

(c) radical function (根式函數)

$$f(x) = x^{1/n} = \sqrt[n]{x} \quad \text{with } n \in \mathbb{N}.$$



Note (根式函數的定義域與值域)

Let $f(x) = \sqrt[n]{x}$ with $n \in \mathbb{N}$.

- When n is odd (奇數), we know that

$$\text{dom}(f) = \text{range}(f) = (-\infty, \infty) = \mathbb{R}.$$

- When n is even (偶數), we see that

$$\text{dom}(f) = \text{range}(f) = [0, \infty).$$



Three Categories of Elementary Functions (2/2)

2. Trigonometric Functions (三角函數)

$$f(x) = \sin x, \cos x, \tan x, \cot x, \sec x, \csc x.$$

3. Exponential and Logarithmic Functions (指數與對數函數)

$$f(x) = a^x \quad \text{or} \quad f(x) = \log_a x,$$

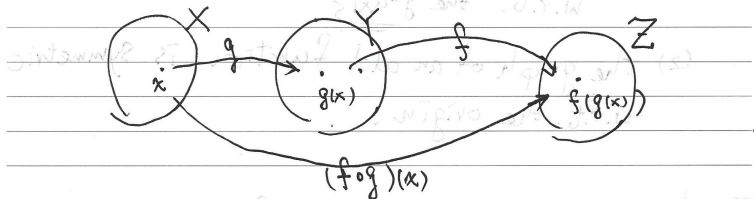
where $0 < a \neq 1$. (See Section 1.4 later!)



Def (合成函數的定義)

Let X , Y and Z be subsets of $\mathbb{R}$. The composite function (合成函數) of $f: Y \rightarrow Z$ and $g: X \rightarrow Y$ is defined by

$$(f \circ g)(x) = f(g(x)) \quad \forall x \in X.$$



Example 4: Let $f(x) = 2x - 3$ and $g(x) = \cos x$.

(a) $(f \circ g)(x) = f(g(x)) = 2 \cos x - 3 \quad \forall x \in \mathbb{R}.$

(b) $(g \circ f)(x) = g(f(x)) = \cos(2x - 3) \quad \forall x \in \mathbb{R}.$

Note: $(f \circ g)(x) \neq (g \circ f)(x)$ in general!

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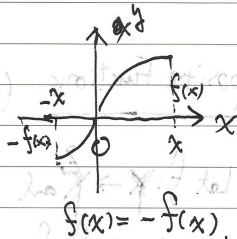
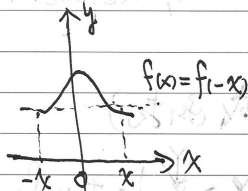


Def (Even and Odd Functions)

Let f be a real-valued function defined on $X \subseteq \mathbb{R}$.

(1) f is an even function (偶函數) if $f(-x) = f(x) \quad \forall x \in X$.

(2) f is an odd function (奇函數) if $f(-x) = -f(x) \quad \forall x \in X$.



Notes:



Notes

- The graph of an even function is symmetric w.r.t. the y -axis.
(偶函數的圖形對稱於 y -軸)
- The graph of an odd function is symmetric w.r.t. the origin.
(奇函數的圖形對稱於原點)



Example 5: Determine whether each function is even, odd or neither.

(a) $f(x) = x^3 - x$ is odd, since $f(-x) = (-x)^3 - (-x)$
 $= -x^3 + x = -(x^3 - x) = -f(x) \quad \forall x \in \mathbb{R}.$

(b) $g(x) = 1 + \cos x$ is even because $g(-x) = 1 + \cos(-x)$
 $= 1 + \cos x = g(x) \quad \forall x \in \mathbb{R}.$

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Section 1.3

Inverse Functions

(反函數)

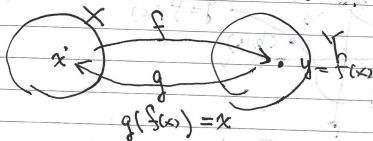


Def (反函數的定義)

Let $f: X \rightarrow Y$ be a function, where $X, Y \subseteq \mathbb{R}$. A function $g: Y \rightarrow X$ is called the inverse function (反函數) of f if

$$g(f(x)) = x \quad \forall x \in X \quad \text{and} \quad f(g(y)) = y \quad \forall y \in Y.$$

In this case, we denote $g = f^{-1}$. (讀作 f -inverse)



Remark: If the inverse function $g \equiv$, then $g = f^{-1}$ is unique.



unique.

Example 1: Show that $f(x) = 2x^3 - 1$ and $g(x) = \sqrt[3]{\frac{x+1}{2}}$ are inverse functions of each other.

Sol: $\because f(g(x)) = 2 \left(\sqrt[3]{\frac{x+1}{2}} \right)^3 - 1 = 2 \left(\frac{x+1}{2} \right) - 1 = x$

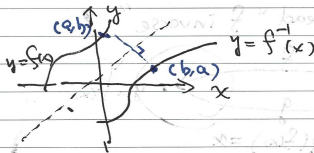
and $g(f(x)) = \sqrt[3]{\frac{2x^3 - 1 + 1}{2}} = \sqrt[3]{x^3} = x \quad \forall x \in \mathbb{R}$.

$\therefore f$ and g are inverse functions of each other.



Remarks

- 1 If $f^{-1} \exists$, then $(f^{-1})^{-1} = f$.
- 2 In general, it is true that $f^{-1}(x) \neq \frac{1}{f(x)}$.
- 3 The graph of f^{-1} is a reflection (反射) of the graph of f in the line $y = x$, i.e., $b = f(a) \iff f^{-1}(b) = a$.



(f 与 f^{-1} 的图形) 反射对称於直线 $y=x$



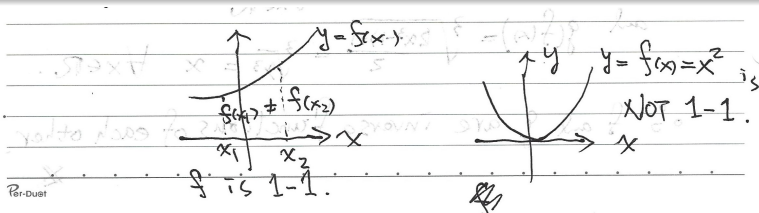
Main Questions

- Does f **always have** an inverse function f^{-1} ?
- When does f^{-1} exist for any real-valued function f ?



Def (一對一函數的定義)

Let f be a real-valued function defined on $X \subseteq \mathbb{R}$. If $f(x_1) \neq f(x_2)$ for any $x_1 \neq x_2 \in X$, then f is an one-to-one function. (一對一函數; 簡寫成 1-1)



Thm (反函數 f^{-1} 的存在性)

Let f be a real-valued function defined on $X \subseteq \mathbb{R}$. Then

$$f^{-1} \text{ } \exists \iff f \text{ is one-to-one on } X.$$



Example 2: (判斷 f^{-1} の存在性).

(a) Let $f(x) = x^3 - 1$. Then f is 1-1 on $\mathbb{R}$.

Reason: suppose that $f(x_1) = f(x_2) \quad \forall x_1, x_2 \in \mathbb{R}$.

$$\Rightarrow x_1^3 - 1 = x_2^3 - 1 \Rightarrow x_1^3 = x_2^3 \Rightarrow x_1 = x_2$$

$\Rightarrow f$ is 1-1 on $\mathbb{R}$.

So, $f(x) = x^3 - 1$ has an inverse function.



(b) Let $f(x) = x^3 - x + 1$. $\forall x \in \mathbb{R}$.

$\Rightarrow f$ is NOT 1-1 on $\mathbb{R}$, since

$$f(-1) = f(0) = f(1) = 1.$$

$\Rightarrow f^{-1} \nexists$ by the above Thm.

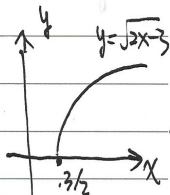


Example 3: Find f^{-1} for $f(x) = \sqrt{2x-3} \quad \forall x \in [\frac{3}{2}, \infty)$.

Sol: $\circ \circ f: [\frac{3}{2}, \infty) \rightarrow [0, \infty)$ is 1-1

$\circ \circ f^{-1}: [0, \infty) \rightarrow [\frac{3}{2}, \infty) \quad \exists !$

$$y = \sqrt{2x-3} \Leftrightarrow y^2 = 2x-3 \Leftrightarrow x = \frac{y^2+3}{2} = f^{-1}(y).$$



Per-Duet

So, the inverse function of f is

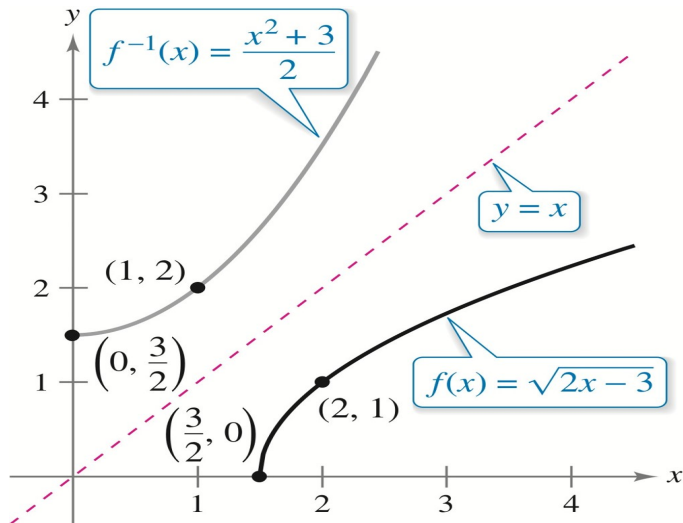
$$f^{-1}(x) = \frac{x^2+3}{2} \quad \forall x \in [0, \infty).$$

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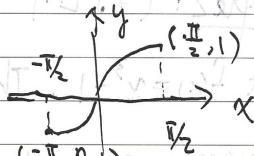
1 1



示意圖 (承上頁)



Example 4: $f(x) = \sin x$ is NOT 1-1 on $\mathbb{R}$ because
 $f(0) = f(\pi) = 0$, but it is 1-1 on
 $[-\frac{\pi}{2}, \frac{\pi}{2}]$.



$$\Rightarrow f^{-1} = \sin^{-1} \neq \arcsin$$

$$= \arcsin : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}] \quad \square !$$



Inverse Trigonometric Functions (反三角函數)

- In order to obtain the inverse trigonometric functions, we need to restrict the domains of six trigonometric functions.
- Conventionally, the following functions

$$\sin : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1], \quad \cos : [0, \pi] \rightarrow [-1, 1]$$

$$\tan : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow (-\infty, \infty), \quad \cot : (0, \pi) \rightarrow (-\infty, \infty)$$

$$\sec : \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right] \rightarrow (-\infty, -1] \cup [1, \infty),$$

$$\csc : \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right] \rightarrow (-\infty, -1] \cup [1, \infty)$$

are both 1-1 on the restricted domains.

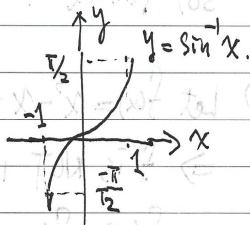


$$① \quad y = \arcsin x = \sin^{-1} x$$

$$\Leftrightarrow \sin y = x$$

$$\text{Domain: } -1 \leq x \leq 1$$

$$\text{Range: } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

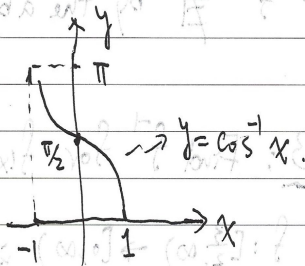


$$② \quad y = \arccos x = \cos^{-1} x$$

$$\Leftrightarrow \cos y = x$$

$$\text{Domain: } -1 \leq x \leq 1$$

$$\text{Range: } 0 \leq y \leq \pi$$

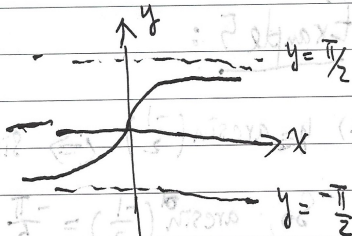


$$(3) \quad y = \arctan x = \tan^{-1} x$$

$$\Leftrightarrow \tan y = x$$

$$\text{Domain: } -\infty < x < \infty$$

$$\text{Range: } -\frac{\pi}{2} < y < \frac{\pi}{2}$$

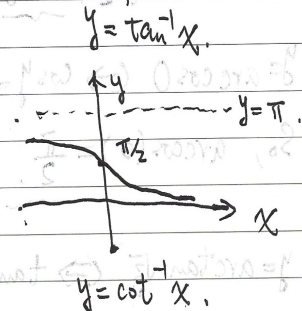


$$(4) \quad y = \operatorname{arccot} x = \cot^{-1} x$$

$$\Leftrightarrow \cot y = x$$

$$\text{Domain: } -\infty < x < \infty$$

$$\text{Range: } 0 < y < \pi$$



$$(5) \quad y = \operatorname{arccsc} x = \csc^{-1} x$$



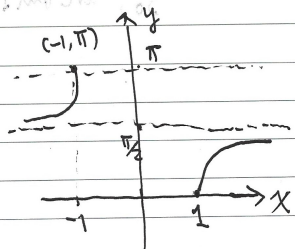
$$(5) y = \operatorname{arcsec} x = \sec^{-1} x$$

$$\Leftrightarrow \sec y = x$$

$$\text{Domain: } |x| \geq 1$$

$$\text{Range: } 0 \leq y \leq \pi, y \neq \frac{\pi}{2}$$

$$y = \cos^{-1} x$$



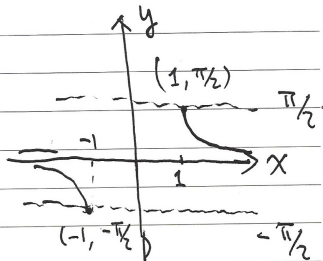
$$(6) y = \operatorname{arccsc} x = \csc^{-1} x$$

$$\Leftrightarrow \csc y = x$$

$$\text{Domain: } |x| \geq 1$$

$$\text{Range: } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, y \neq 0$$

$$y = \sec^{-1} x$$



Example 5 :

$$(a) \ y = \arcsin\left(-\frac{1}{2}\right) \Leftrightarrow \sin y = -\frac{1}{2} \Leftrightarrow y = -\frac{\pi}{6}$$

$$\text{So, } \arcsin\left(-\frac{1}{2}\right) = \underline{-\frac{\pi}{6}} \quad \times$$

$$(b) \ y = \arccos 0 \Leftrightarrow \cos y = 0 \Leftrightarrow y = \frac{\pi}{2}$$

$$\text{So, } \arccos(0) = \underline{\frac{\pi}{2}} \quad \times$$

$$(c) \ y = \arctan \sqrt{3} \Leftrightarrow \tan y = \sqrt{3} \Leftrightarrow y = \frac{\pi}{3}$$

$$\text{So, } \arctan \sqrt{3} = \underline{\frac{\pi}{3}} \quad \times$$



Section 1.4

Exponential and Logarithmic Functions

(指數函數與對數函數)



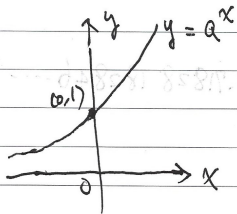
Def (以 a 為底的指數函數)

The exponential function with base number a is defined by

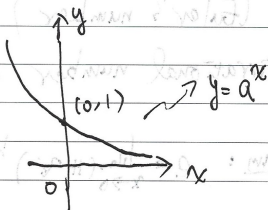
$$f(x) = a^x \quad \forall x \in \mathbb{R},$$

where $0 < a \neq 1$.





$$a > 1$$



$$0 < a < 1$$

Example 2: Sketch the graphs of $y = 2^x$, $y = (\frac{1}{2})^x$ and $y = 3^x$.

Sol:

x	-3	-2	-1	0	1	2	3	4
2^x	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8	16
2^{-x}	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$
3^x	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27	81

Per-Duet



Thm (Basic Properties of a^x)

Let $f(x) = a^x$ with $0 < a \neq 1$. Then

- 1 $\text{dom}(f) = \mathbb{R} = (-\infty, \infty)$.
- 2 $\text{range}(f) = (0, \infty)$, i.e., $f(x) = a^x > 0 \quad \forall x \in \mathbb{R}$.
- 3 $f(0) = a^0 = 1$, i.e., the y -intercept of f is $(0, 1)$, and $f(1) = a$.
- 4 f is one-to-one on $\mathbb{R}$, i.e., $f^{-1} : (0, \infty) \rightarrow \mathbb{R} \quad \exists$.



Thm (Laws of Exponents; 指數律)

If $a, b > 0$ and $x, y \in \mathbb{R}$, then

- ① $a^x a^y = a^{x+y}.$
- ② $(a^x)^y = a^{xy} = (a^y)^x.$
- ③ $(ab)^x = a^x b^x.$
- ④ $\frac{a^x}{a^y} = a^{x-y}.$
- ⑤ $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}.$



Def (Euler's number; 歐拉數或尤拉數)

The irrational number $e = \lim_{x \rightarrow 0} (1 + x)^{1/x} \approx 2.71828182846 \dots$.

Note: see Example 3 for observing the behavior of $f(x) = (1 + x)^{1/x}$ as x approaches 0.

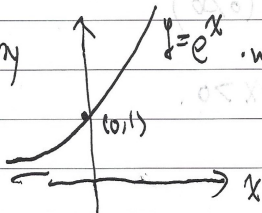
Definition of e^x

For the base number $a = e > 1$, $f(x) = a^x = e^x$ is called the natural exponential function (自然指數函數).



Example 4: The graph of $f(x) = e^x$ $x \in \mathbb{R}$ is

given by $y = e^x$ with $e > 1$.



The Inverse Function of e^x

Since the function $f(x) = e^x$ is one-to-one on $\mathbb{R}$, it must have an inverse function $f^{-1} : (0, \infty) \rightarrow \mathbb{R} = (-\infty, \infty)$!

Definition of $\ln x$

The inverse function of $f(x) = e^x$, denoted by $f^{-1}(x) = \ln x$, is called the natural logarithmic function (自然對數函數). Moreover, we have

$$y = \ln x \quad \forall x > 0 \iff e^y = x \quad \forall y \in \mathbb{R}.$$



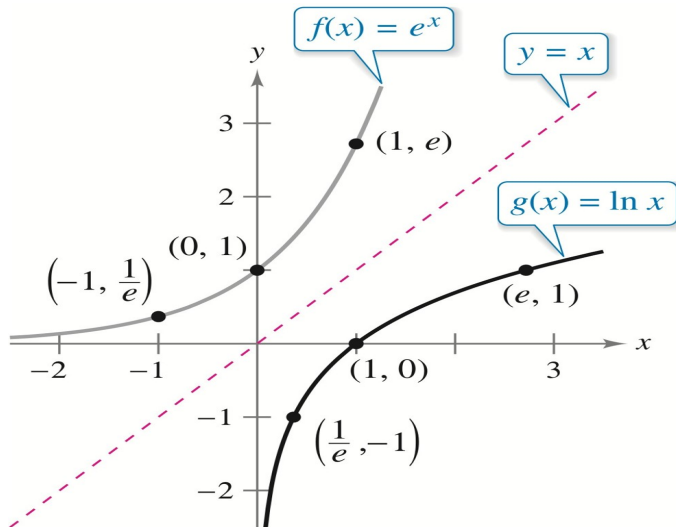
Thm (Basic Properties of $\ln x$)

Let $f(x) = \ln x$. Then

- 1 $\text{dom}(f) = (0, \infty)$.
- 2 $\text{range}(f) = \mathbb{R} = (-\infty, \infty)$.
- 3 $f(1) = \ln 1 = 0$, i.e., the x -intercept of f is $(1, 0)$, and $f(e) = \ln e = 1$.
- 4 f is one-to-one on $(0, \infty)$, i.e., $f^{-1} : \mathbb{R} \rightarrow (0, \infty) \quad \exists$.
- 5 $\ln(e^x) = x$ for $x \in \mathbb{R}$, and $e^{\ln x} = x$ for $x > 0$.



示意圖 (承上頁)



Thm (Laws of Logarithms; 對數律)

If $x > 0$, $y > 0$ and $z \in \mathbb{R}$, then

① $\ln(xy) = \ln x + \ln y.$

② $\ln\left(\frac{x}{y}\right) = \ln x - \ln y.$

③ $\ln(x^z) = z \cdot \ln x.$



Example 5 :

(a), (b), (c) 自行閱讀.

$$\begin{aligned} \text{(d)} \quad \ln \frac{(x^2+3)^2}{x \sqrt{x^2+1}} &= \ln[(x^2+3)^2] - \ln[x \sqrt{x^2+1}] \\ &= 2 \ln(x^2+3) - \ln x - \frac{1}{2} \ln(x^2+1). \end{aligned}$$



Example 6: Solve the equations.

(a) $7 = e^{x+1} \Rightarrow \ln 7 = (x+1) \ln e = x+1 \Rightarrow x = \ln 7 - 1$

(b) $\ln(2x-3) = 5 \Rightarrow 2x-3 = e^5 \Rightarrow x = \frac{1}{2}(e^5+3)$



Section 1.5

Finding Limits Graphically and Numerically

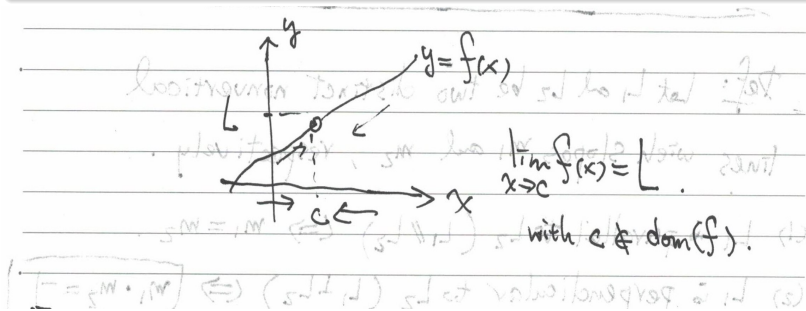
(從圖形上與數值上求極限)



Informal Definition of a Limit

Def (函數極限值的非正式定義)

Let f be a real-valued function defined on $X \subseteq \mathbb{R}$ with $c \notin X$. If $f(x)$ becomes **arbitrarily close** to a unique number $L \in \mathbb{R}$ as x approaches c **from either side**, then the limit of f is L as x approaches c , denoted by $\lim_{x \rightarrow c} f(x) = L$.



Example 1: Find $\lim_{x \rightarrow 0} \frac{x}{\sqrt{x+1} - 1}$ numerically.

Sol:

Let $f(x) = \frac{x}{\sqrt{x+1} - 1}$ for $x \neq 0$.

Note that $\text{dom}(f) = (-\infty, 0) \cup (0, \infty)$ and $c = 0 \notin \text{dom}(f)$.

x	-10^{-2}	-10^{-3}	-10^{-4}	0	10^{-4}	10^{-3}	10^{-2}
$f(x)$			1.99995	?	2.00005		

$\Rightarrow \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{x}{\sqrt{x+1} - 1} = 2$

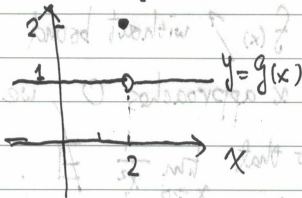
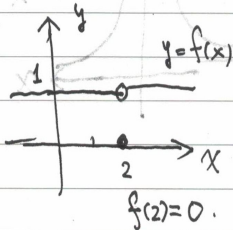
Per-Duest



Example 2:

$$\text{if } f(x) = \begin{cases} 1, & x \neq 2 \\ 0, & x = 2 \end{cases} \text{ and } g(x) = \begin{cases} 1, & x \neq 2 \\ 2, & x = 2 \end{cases}$$

$$\text{then } \lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} g(x) = 1 \quad g(2) = 2$$



Type I (在 c 點兩邊的極限值不相等)

If $f(x) \rightarrow L_1$ and $f(x) \rightarrow L_2$, where $L_1 \neq L_2$, as x approaches c from either side, then $\lim_{x \rightarrow c} f(x) \nexists$.



Example 3: Show that $\lim_{x \rightarrow 0} \frac{|x|}{x} \nexists$.

Sol: Let $f(x) = \frac{|x|}{x} = \begin{cases} \frac{x}{x} = 1 & \text{if } x > 0 \\ \frac{-x}{x} = -1 & \text{if } x < 0 \end{cases}$.

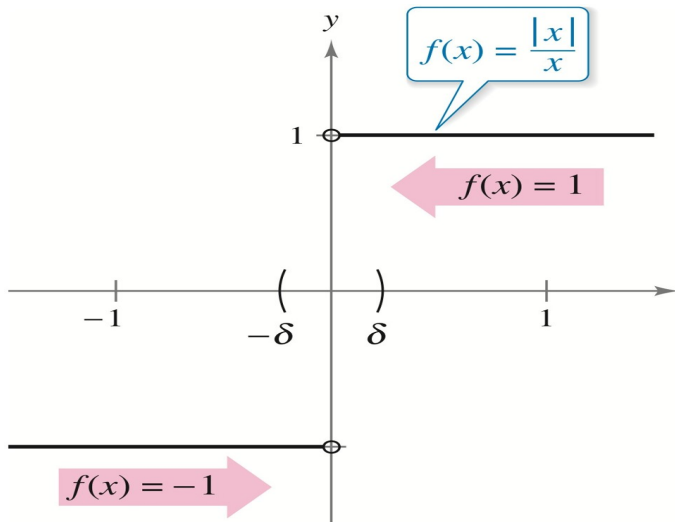
$\Rightarrow f(x) \rightarrow 1$ as $x \rightarrow 0$ from the right side.

$f(x) \rightarrow -1$ as $x \rightarrow 0$ from the left side.

$\Rightarrow \lim_{x \rightarrow 0} \frac{|x|}{x} \nexists$.



示意圖 (承上例)



Type II (函數值在 c 點附近無上下界)

If $f(x)$ increases (遞增) or decreases (遞減) without bound as x approaches c from either side, then $\lim_{x \rightarrow c} f(x) \nexists$.



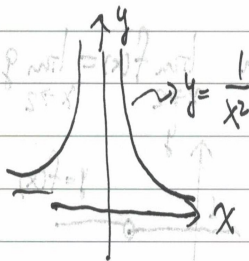
Example 4 : $f(x)$ becomes unbounded

Consider $f(x) = \frac{1}{x^2}$ for $x \neq 0$.

Since $f(x) \nearrow$ without bound
as x approaches 0 , we

know that $\lim_{x \rightarrow 0} \frac{1}{x^2} \nexists$.

Fact: $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$.



Type III (函數值在 c 點附近震盪)

If $f(x)$ oscillates (震盪) between two fixed values as x approaches c from either side, then $\lim_{x \rightarrow c} f(x) \nexists$.

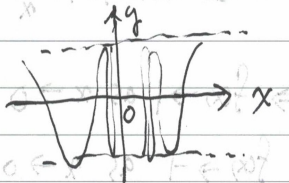


Example 5: ($\sin$ is oscillating)

Find the limit $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$.

Sol: $\because f(x) = \sin\left(\frac{1}{x}\right)$ oscillates between -1 and 1 as x approaches 0 .

$\therefore \lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right) \nexists$

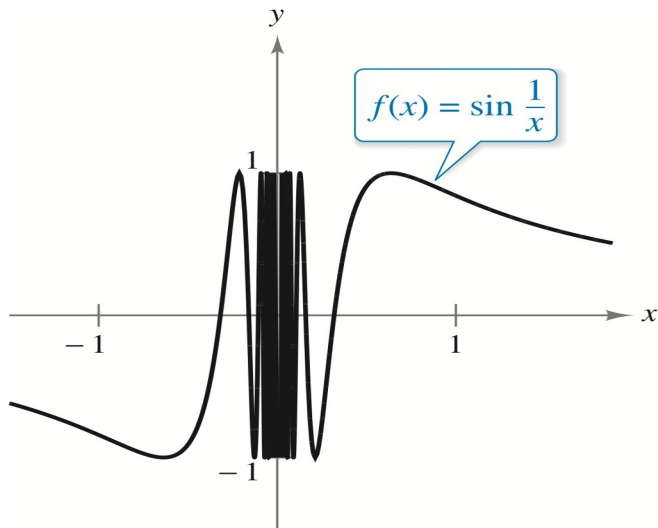


Facts: ① For $x = \frac{2}{(4n+1)\pi}$, $\sin\left(\frac{1}{x}\right) = \sin\left(\frac{(4n+1)\pi}{2}\right) = 1$.

② For $x = \frac{2}{(4n-1)\pi}$, $\sin\left(\frac{1}{x}\right) = -1$.



示意圖 (承上例)



Note: 函數值在 -1 和 1 之間來回 [振盪]，但沒碰到 y -軸喔!



Def (函數極限值的正式定義)

Let f be a real-valued function defined on $X \subseteq \mathbb{R}$ with $c \notin X$. Then

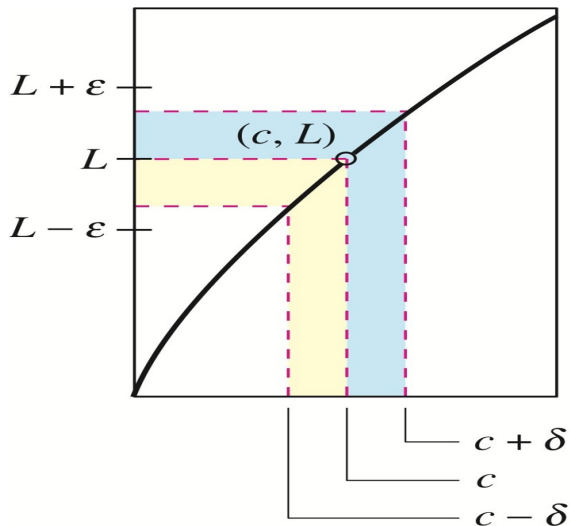
$$\lim_{x \rightarrow c} f(x) = L.$$

$$\iff \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. if } 0 < |x - c| < \delta \text{ (and } x \in X), \\ \text{then } |f(x) - L| < \varepsilon.$$

Note: this is also called the ε - δ definition of a limit.



極限的正式定義



Example 6: Find a $\delta > 0$ s.t. if $0 < |x-3| < \delta$,

then $|(2x-5)-1| < \varepsilon = 0.01$.

Sol: Want $|(2x-5)-1| = |2x-6| = 2|x-3| < 0.01$

$$\Rightarrow |x-3| < \frac{1}{2}(0.01) = 0.005 = \delta$$

So, we may choose $\delta = 0.005$.

Remark: Any $\delta \in (0, 0.005]$ also works in this example!



Example 7: Show that $\lim_{x \rightarrow 2} (3x-2) = 4$.

If: Let $f(x) = 3x-2$ and $L = 4$.

Given $\varepsilon > 0$ arbitrarily.

Choose $\delta = \frac{\varepsilon}{3} > 0$.

If $0 < |x-2| < \delta$, then $|f(x) - L| = |(3x-2) - 4| = 3|x-2|$
 $< 3\delta = 3\left(\frac{\varepsilon}{3}\right) = \varepsilon$.

Thus, $\lim_{x \rightarrow 2} (3x-2) = 4$ by the ε - δ Def.



Section 1.6

Evaluating Limits Analytically

(從分析上求極限)



Thm (Basic Limit Laws; 1/2)

Let $b, c \in \mathbb{R}$ and let f and g be real-valued functions with

$$\lim_{x \rightarrow c} f(x) = L, \quad \lim_{x \rightarrow c} g(x) = K.$$

$$(1) \quad \lim_{x \rightarrow c} b = b \text{ and } \lim_{x \rightarrow c} x = c.$$

$$(2) \quad \lim_{x \rightarrow c} x^n = c^n \text{ and } \lim_{x \rightarrow c} [f(x)]^n = L^n \quad \forall n \in \mathbb{N}.$$

$$(3) \quad \lim_{x \rightarrow c} [b \cdot f(x)] = b \cdot \left[\lim_{x \rightarrow c} f(x) \right] = b \cdot L.$$

$$(4) \quad \lim_{x \rightarrow c} [f(x) \pm g(x)] = \left[\lim_{x \rightarrow c} f(x) \right] \pm \left[\lim_{x \rightarrow c} g(x) \right] = L \pm K.$$



Thm (Basic Limit Laws; 2/2)

$$(5) \lim_{x \rightarrow c} [f(x) \cdot g(x)] = \left[\lim_{x \rightarrow c} f(x) \right] \cdot \left[\lim_{x \rightarrow c} g(x) \right] = L \cdot K.$$

$$(6) \lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)} = \frac{L}{K} \text{ if } K \neq 0.$$

(7) The Limit of $f \circ g$: (合成函數的極限值)

If $\lim_{x \rightarrow c} f(x) = f(K)$, then $\lim_{x \rightarrow c} f(g(x)) = f\left(\lim_{x \rightarrow c} g(x)\right) = f(K)$.



Examples 1 and 2: $\frac{5+x+x^2}{1+x}$ and $\lim_{x \rightarrow 2} \frac{5+x+x^2}{1+x}$

(a) $\lim_{x \rightarrow 2} 3 = 3$ $\frac{5}{5} = \frac{5+1+1}{1+1} = \frac{5+x+x^2}{1+x}$ and $\lim_{x \rightarrow 2} \frac{5+x+x^2}{1+x}$

(b) $\lim_{x \rightarrow -4} x = -4$

(c) $\lim_{x \rightarrow 2} x^2 = (2)^2 = 4$

(d) $\lim_{x \rightarrow 2} (4x^2 + 3) = 4 \left(\lim_{x \rightarrow 2} x^2 \right) + \lim_{x \rightarrow 2} 3 = 4(2^2) + 3 = 19$



Thm (Limits of Elementary Functions; 1/2)

Let c be a real number in the domain of the given function.

- (1) If $p(x)$ is a polynomial, then $\lim_{x \rightarrow c} p(x) = p(c)$.
- (2) If $r(x) = p(x)/q(x)$ is a rational function with $q(c) \neq 0$, then $\lim_{x \rightarrow c} r(x) = r(c) = p(c)/q(c)$.
- (3) $\lim_{x \rightarrow c} \sqrt[n]{x} = \sqrt[n]{c} \quad \forall n \in \mathbb{N}$, where $c \geq 0$ when n is even and $c \in \mathbb{R}$ when n is odd.



Thm (Limits of Elementary Functions; 2/2)

(4) Limits of 6 trigonometric functions are given by

$$\begin{aligned}\lim_{x \rightarrow c} \sin x &= \sin c, & \lim_{x \rightarrow c} \cos x &= \cos c, & \lim_{x \rightarrow c} \tan x &= \tan c, \\ \lim_{x \rightarrow c} \cot x &= \cot c, & \lim_{x \rightarrow c} \sec x &= \sec c, & \lim_{x \rightarrow c} \csc x &= \csc c.\end{aligned}$$

(5) $\lim_{x \rightarrow c} a^x = a^c$ for $a > 0$ and $c \in \mathbb{R}$.

(6) $\lim_{x \rightarrow c} \ln x = \ln c$ for $c > 0$.



Example 3: Find $\lim_{x \rightarrow 1} \frac{x^2 + x + 2}{x + 1}$

Sol. $\lim_{x \rightarrow 1} \frac{x^2 + x + 2}{x + 1} = \frac{1 + 1 + 2}{1 + 1} = \frac{4}{2} = 2$

Example 4: (根号函数的极限)

(a)

$$\lim_{x \rightarrow 0} \sqrt{x^2 + 4} = \sqrt{\lim_{x \rightarrow 0} (x^2 + 4)} = \sqrt{0 + 4} = 2$$

(b)

$$\lim_{x \rightarrow 3} \sqrt[3]{2x^2 - 10} = \sqrt[3]{18 - 10} = \sqrt[3]{8} = 2$$



Example 5 : (Limits of Transcendental Functions)

$$(a) \lim_{x \rightarrow 0} \tan x = \tan(0) = 0.$$

$$(b) \lim_{x \rightarrow 0} \sin^2 x = \left(\lim_{x \rightarrow 0} \sin x \right)^2 = 0^2 = 0.$$

$$(c) \lim_{x \rightarrow -1} x e^x = \left(\lim_{x \rightarrow -1} x \right) \left(\lim_{x \rightarrow -1} e^x \right) = \underline{-e^{-1}}.$$

$$(d) \lim_{x \rightarrow e} \ln x^3 = \left(\lim_{x \rightarrow e} 3 \ln x \right) = \underline{(3)(1) = 3}.$$



Thm 1.7 (化簡函數後求極限值)

If $\exists \delta > 0$ s.t. $f(x) = g(x) \quad \forall x \in (c - \delta, c) \cup (c, c + \delta)$, then

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x).$$

Note: 將 $f(x)$ 簡化為 $g(x)$ 後，兩者在 c 點附近的極限值相等!



Example 7: (Dividing Out Technique)

$$\lim_{x \rightarrow (-3)} \frac{x^2 + x - 6}{x + 3} = \lim_{x \rightarrow (-3)} \frac{(x+3)(x-2)}{x+3} = \lim_{x \rightarrow (-3)} (x-2)$$
$$= -3 - 2 = -5$$

有理化技巧.

Example 8: (Rationalizing Technique)

有理化技巧.

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x} = \lim_{x \rightarrow 0} \left(\frac{\sqrt{x+1} - 1}{x} \times \frac{\sqrt{x+1} + 1}{\sqrt{x+1} + 1} \right)$$
$$= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+1} + 1)} = \frac{1}{\sqrt{0+1} + 1} = \frac{1}{2}$$



Thm 1.8 (Squeeze (or Sandwich) Thm; 夾擠定理或夾擊定理)

If $\exists \delta > 0$ s.t. $h(x) \leq f(x) \leq g(x) \quad \forall x \in (c - \delta, c) \cup (c, c + \delta)$, and

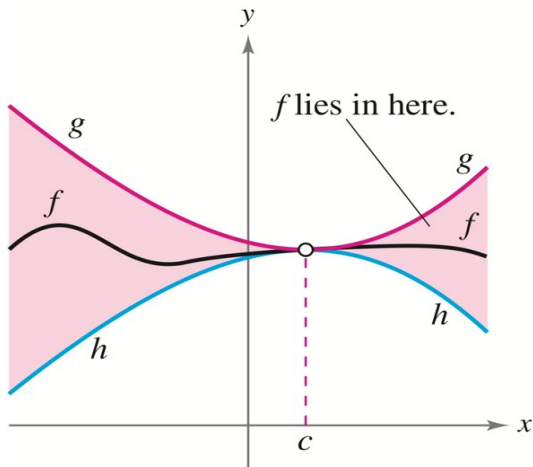
$$\lim_{x \rightarrow c} h(x) = L = \lim_{x \rightarrow c} g(x),$$

then $\lim_{x \rightarrow c} f(x) = L$.



Thm 1.8 的示意圖

$$h(x) \leq f(x) \leq g(x)$$



Thm 1.9 (Some Special Limits)

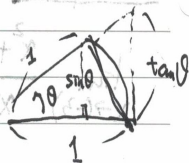
- ① $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$
- ② $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0.$
- ③ $\lim_{x \rightarrow 0} (1 + x)^{1/x} = e.$



pf: ① Note that for $0 < \theta < \frac{\pi}{2}$, we have

$$\frac{\sin \theta}{2} < \frac{\theta}{2} < \frac{\tan \theta}{2}$$

$$\Rightarrow 1 < \frac{\theta}{\sin \theta} < \frac{1}{\cos \theta} \quad \left(\text{Divide by } \frac{\sin \theta}{2} \right)$$



$$\Rightarrow \cos \theta < \frac{\sin \theta}{\theta} < 1 \quad \forall \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \text{ since}$$

$$\cos \theta = \cos(-\theta) \text{ and } \frac{\sin \theta}{\theta} = \frac{\sin(-\theta)}{-\theta} \text{ are even.}$$

From the Squeeze Thm and $\lim_{\theta \rightarrow 0} \cos \theta = \lim_{\theta \rightarrow 0} 1 = 1 \Rightarrow$

$$\boxed{\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1}$$



$$\begin{aligned}
 \textcircled{2} \quad \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} &= \lim_{\theta \rightarrow 0} \left(\frac{1 - \cos \theta}{\theta} \times \frac{1 + \cos \theta}{1 + \cos \theta} \right) \\
 &= \lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta(1 + \cos \theta)} = \left(\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \right) \left(\lim_{\theta \rightarrow 0} \frac{\sin \theta}{1 + \cos \theta} \right) = (1)(0) \\
 &= 0
 \end{aligned}$$



Example 9: $\lim_{x \rightarrow 0} \frac{\tan x}{x} = \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \left(\lim_{x \rightarrow 0} \frac{1}{\cos x} \right) = (1)(1) = 1$

Example 10: Find $\lim_{x \rightarrow 0} \frac{\sin 4x}{x}$.

Sol: Let $\theta = 4x$. Then $\theta \rightarrow 0$ as $x \rightarrow 0$.

$$\begin{aligned} \text{So, } \lim_{x \rightarrow 0} \frac{\sin 4x}{x} &= \lim_{x \rightarrow 0} \left(\frac{\sin 4x}{4x} \right) \times 4 = 4 \left(\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \right) \\ &= (4)(1) = 4. \end{aligned}$$



Section 1.7

Continuity and One-Sided Limits

(連續性與單邊極限)



Def (實值函數的連續性)

Let f be a real-valued function defined on $I = (a, b)$ with $c \in I$.

- (1) f is continuous (連續的; 簡寫為 conti.) at c if $\lim_{x \rightarrow c} f(x) = f(c)$.
- (2) f is conti. on I if it is conti. at each $c \in I$.
- (3) f is everywhere conti. (處處連續) if it is conti. on $\mathbb{R} = (-\infty, \infty)$.



Def (函數的不連續性)

Let f be a real-valued function defined on $I = (a, b)$ with $c \in I$.

- (1) f has a discontinuity (不連續點; 簡寫為 *disconti.*) at c if it is NOT conti. at c .
- (2) A *disconti.* of f at c is called **removable** (可移除的) if f can be made conti, at c **by redefining $f(c)$** . Otherwise, the *disconti.* at c is called **nonremovable** (不可移除的).



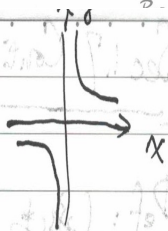
Example 1: Discuss the continuity of each function.

(a) $f(x) = \frac{1}{x}$ is conti. on $(-\infty, 0) \cup (0, \infty) = \text{dom}(f)$, since ~~for all~~

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \frac{1}{x} = \frac{1}{c} \text{ for } c \neq 0.$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x} \nexists$$

The disconti. at $x=0$ is nonremovable.



(b) $f(x) = \frac{x^2-1}{x-1}$ is conti. on $(-\infty, 1) \cup (1, \infty) = \text{dom}(f)$. $f(x) = \frac{1}{x} \quad \forall x \neq 0$

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{(x+1)(x-1)}{x-1} = \lim_{x \rightarrow 1} (x+1) = 2$$

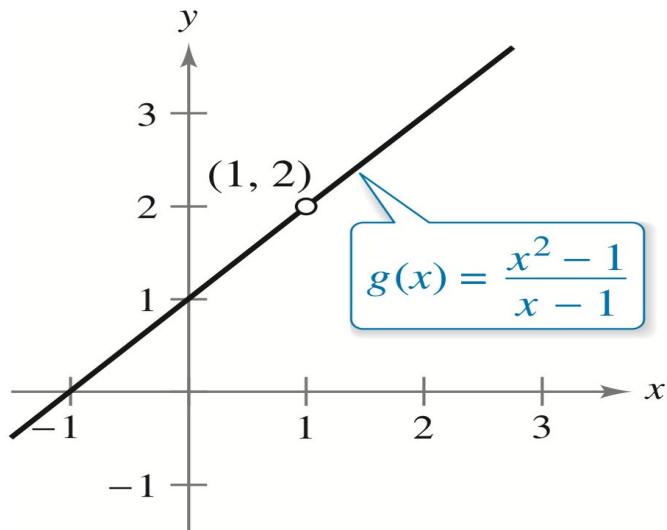
The disconti. at $x=1$ is removable by redefining

$$f(1) = 2 = \lim_{x \rightarrow 1} f(x)$$

$$\Rightarrow \tilde{f}(x) = \begin{cases} \frac{x^2-1}{x-1} = x+1 & \text{for } x \neq 1 \\ 2 & \text{for } x=1 \end{cases} \text{ is conti. on } \mathbb{R}.$$



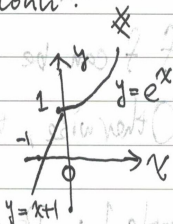
示意圖 (承上例)



(c)
$$h(x) = \begin{cases} x+1, & x \leq 0 \\ e^x, & x > 0 \end{cases}$$
 is conti. on $(-\infty, 0)$ and $(0, \infty)$.

Since $\lim_{x \rightarrow 0} h(x) = 1 = h(0)$, h is conti. at $x=0$.

$\Rightarrow h$ is ~~everywhere~~ everywhere conti.



(d) $y = \sin x$ is everywhere conti.

Since $\lim_{x \rightarrow c} \sin x = \sin c \quad \forall c \in \mathbb{R}$.

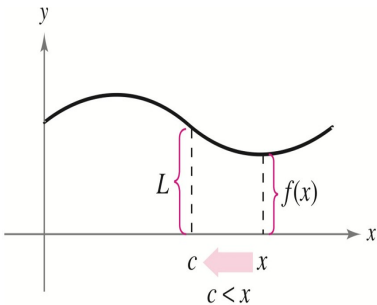
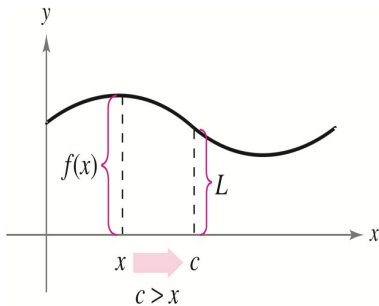


Def (單邊極限值的定義)

- (1) f has the limit L from the right (or the right-hand limit L ; 右極限值) at c , denoted by $\lim_{x \rightarrow c^+} f(x) = L$, if $f(x) \rightarrow L$ as $x \rightarrow c$ from the right.
- (2) f has the limit L from the left (or the left-hand limit L ; 左極限值) at c , denoted by $\lim_{x \rightarrow c^-} f(x) = L$, if $f(x) \rightarrow L$ as $x \rightarrow c$ from the left.



單邊極限值的示意圖



✱
✱✱

Example 3 (最大整数函数)

The greatest integer function is defined by

$$f(x) = \lfloor x \rfloor = \text{the greatest } n \in \mathbb{Z} \text{ s.t. } n \leq x.$$



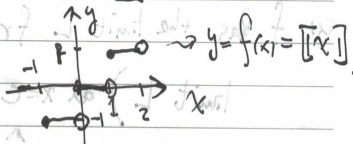
Solution of Example 3

$$\lim_{x \rightarrow n^+} \lfloor x \rfloor = n \text{ and } \lim_{x \rightarrow n^-} \lfloor x \rfloor = n-1 \quad \forall n \in \mathbb{Z}.$$

$$\lim_{x \rightarrow n} \lfloor x \rfloor \nexists$$

$\Rightarrow f(x) = \lfloor x \rfloor$ has a nonremovable

disconti. at each $x = n \in \mathbb{Z}$. $\#$



Thm 1.10 (函數極限值存在的等價條件)

$$\lim_{x \rightarrow c} f(x) = L \iff \lim_{x \rightarrow c^-} f(x) = L = \lim_{x \rightarrow c^+} f(x).$$

(f 在 c 點的極限值為 $L \iff$ 左右極限值均為 L)



One-Sided Continuity

Def (單邊連續的定義)

Let f be a real-valued function defined on $X \subseteq \mathbb{R}$ with $c \in X$.

(1) f is conti. from the right (右連續) at c if $\lim_{x \rightarrow c^+} f(x) = f(c)$.

(2) f is conti. from the left (左連續) at c if $\lim_{x \rightarrow c^-} f(x) = f(c)$.

Remark

f is conti. at $c \iff f$ is conti. from the right and from the left at c .
(f 在 c 點連續 $\iff f$ 在 c 點右連續且左連續)



Continuity on a Closed Interval

Def (在閉區間上的連續性)

f is conti. on $I = [a, b]$ if the following conditions hold:

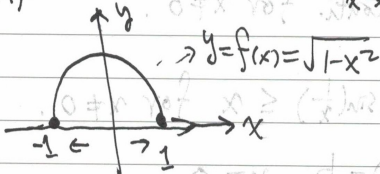
- 1 f is conti. on the open interval (a, b) .
- 2 f is conti. from the right at a , i.e., $\lim_{x \rightarrow a^+} f(x) = f(a)$.
- 3 f is conti. from the left at b , i.e., $\lim_{x \rightarrow b^-} f(x) = f(b)$.



Example 4: $f(x) = \sqrt{1-x^2}$ is conti. on $I = [-1, 1] = \text{dom}(f)$

because f is conti. on $(-1, 1)$, and

$$\lim_{x \rightarrow (-1)^+} \sqrt{1-x^2} = 0 = f(-1) \text{ and } \lim_{x \rightarrow 1^-} \sqrt{1-x^2} = 0 = f(1).$$



Thm (連續函數的性質)

- 1 If f and g are conti. at c and $b \in \mathbb{R}$, then $f \pm g$, bf , fg and f/g with $g(c) \neq 0$ are conti. at c , respectively.
- 2 If g is conti. at c and f is conti. at $g(c)$, then $(f \circ g)(x) = f(g(x))$ is conti. at c .
- 3 All elementary functions are conti. on their domains, including one-sided continuity!



Example 7 :

(a) $f(x) = \tan x$ is conti. on $\text{dom}(f) = \{x \in \mathbb{R} \mid x \neq (n + \frac{1}{2})\pi, n \in \mathbb{Z}\}$

(b) $g(x) = \begin{cases} \sin(\frac{1}{x}), & x \neq 0 \\ 0, & x = 0 \end{cases}$ is conti. on $(-\infty, 0)$ and $(0, \infty)$

because $\frac{1}{x}$ is conti. for $x \neq 0$ and $\sin x$ is conti. on $\mathbb{R}$.

Since $\lim_{x \rightarrow 0} g(x) \nexists$, g has a (non removable) disconti.

at $x=0$.



(c) $h(x) = \begin{cases} x \sin(\frac{1}{x}), & x \neq 0 \\ 0, & x = 0 \end{cases}$ is everywhere conti.

Reason: ① h is conti. for $x \neq 0$.

② $\because -|x| \leq x \sin(\frac{1}{x}) \leq |x|$ for $x \neq 0$.

$$\text{and } \lim_{x \rightarrow 0} (-|x|) = \lim_{x \rightarrow 0} |x| = 0$$

$$\therefore \lim_{x \rightarrow 0} h(x) = \lim_{x \rightarrow 0} x \sin(\frac{1}{x}) = 0 \text{ by the Squeeze Thm.} \\ = h(0)$$

$\Rightarrow h$ is conti. at $x=0$.

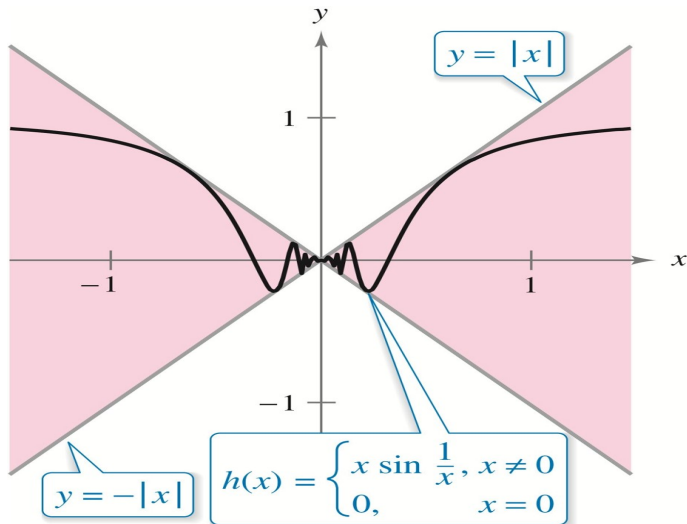
From ① and ② $\Rightarrow h$ is conti. on $\mathbb{R}$. $\#$

Thm:

$$\lim_{x \rightarrow c} f(x) = L \Leftrightarrow \lim_{x \rightarrow c} |f(x) - L| = 0 \Leftrightarrow \lim_{x \rightarrow c} |f(x)| = |L|.$$



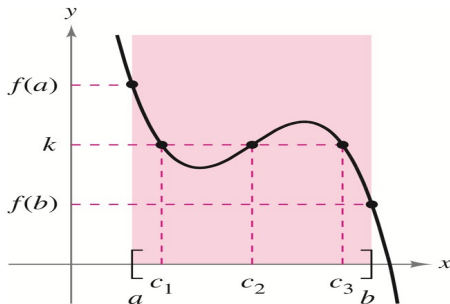
示意圖 (承上例)



Intermediate Value Theorem

Thm 1.13 (I.V.T.; 中間值定理)

If f is **conti.** on $[a, b]$, $f(a) \neq f(b)$ and k is any number between $f(a)$ and $f(b)$, then $\exists c \in [a, b]$ s.t. $f(c) = k$.



Example 8: (Application of I.V.T.)

Show that $f(x) = x^3 + 2x - 1$ has a zero in $[0, 1]$.

pf: $\because f$ is conti. on $[0, 1]$ and $f(0) = -1 < 0 < 2 = f(1)$

$\therefore$ By I.V.T. $\Rightarrow \exists c \in (0, 1)$ s.t. $f(c) = 0$.

So, f has a zero c in $[0, 1]$.



Section 1.8

Infinite Limits

(無窮極限)

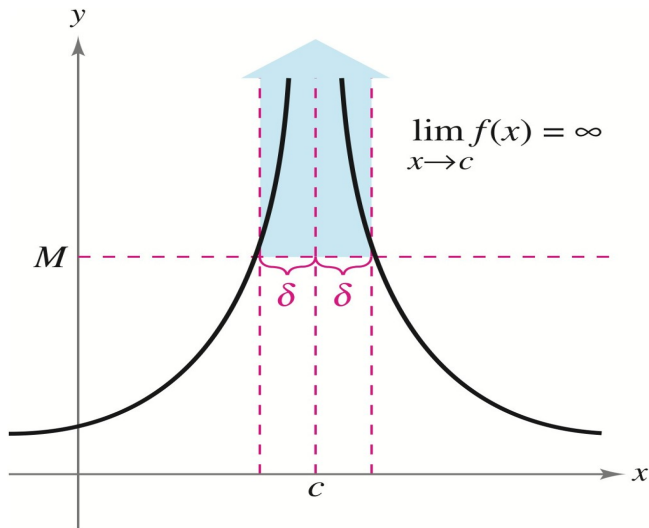


Def (無窮極限值的定義; 1/2)

- (1) $\lim_{x \rightarrow c} f(x) = \infty \iff \forall M > 0, \exists \delta > 0 \text{ s.t. if } 0 < |x - c| < \delta,$
then $f(x) > M$.
- (2) $\lim_{x \rightarrow c} f(x) = -\infty \iff \forall N < 0, \exists \delta > 0 \text{ s.t. if } 0 < |x - c| < \delta,$
then $f(x) < N$.



示意圖 (承上頁)



Def (無窮極限值的定義; 2/2)

$$(3) \lim_{x \rightarrow c^+} f(x) = \infty \left(\text{or } \lim_{x \rightarrow c^-} f(x) = \infty \right) \iff \forall M > 0, \exists \delta > 0$$

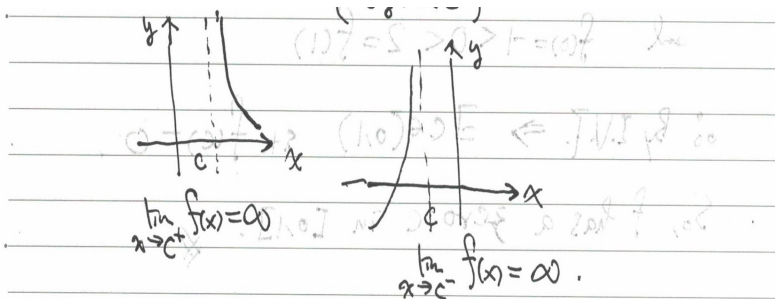
s.t. if $c < x < c + \delta$ (or $c - \delta < x < c$), then $f(x) > M$.

$$(4) \lim_{x \rightarrow c^+} f(x) = -\infty \left(\text{or } \lim_{x \rightarrow c^-} f(x) = -\infty \right) \iff \forall N < 0, \exists \delta > 0$$

s.t. if $c < x < c + \delta$ (or $c - \delta < x < c$), then $f(x) < N$.



示意圖 (承上頁)



重要口訣 (切記!)

若 $+0$ 與 -0 分別表示接近零的極小正數與負數, 則

- $\frac{1}{+0} = \infty, \quad \frac{1}{-0} = -\infty$

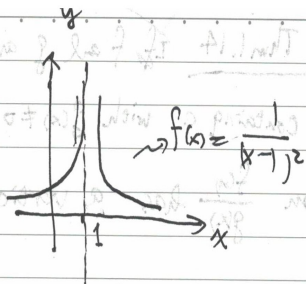
- $\frac{1}{\infty} = \frac{1}{-\infty} = 0,$

其中, $+\infty = \infty$ 和 $-\infty$ 分別為正負無窮遠處的符號。



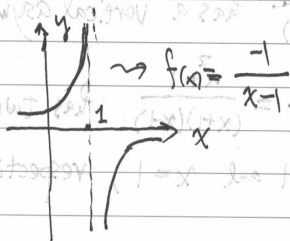
Example 1:

(a) $\lim_{x \rightarrow 1} \frac{1}{(x-1)^2} = \infty$



(b) If $f(x) = \frac{-1}{x-1}$ for $x \neq 1$,

then $\lim_{x \rightarrow 1^+} \frac{-1}{x-1} = -\infty$ and $\lim_{x \rightarrow 1^-} \frac{-1}{x-1} = \infty$



Example 4: Find the limits.

$$(a) \lim_{x \rightarrow 1^-} \frac{x^2 - 3x}{x - 1} \left(= \frac{-2}{-0} \right) = \infty$$

$$(b) \lim_{x \rightarrow 1^+} \frac{x^2 - 3x}{x - 1} \left(= \frac{-2}{+0} \right) = -\infty$$



Vertical Asymptotes

Def (鉛直漸近線或垂直漸近線)

If $\lim_{x \rightarrow c^+} f(x) = \pm\infty$ or $\lim_{x \rightarrow c^-} f(x) = \pm\infty$, then the line $x = c$ is a vertical asymptote (垂直漸近線) of the graph of f .

Thm 1.14 (判斷垂直漸近線的位置)

If f and g are conti. on an open interval I containing c , where $g(x) \neq 0 \quad \forall x \in I \setminus \{c\}$. If $f(c) \neq 0$ and $g(c) = 0$, then $\frac{f(x)}{g(x)}$ has a vertical asymptote at $x = c$.



Example 2: (Finding Vertical asymptotes)

(a) $f(x) = \frac{1}{2(x+1)}$ has a vertical asymptote at $x = -1$.

(b) $f(x) = \frac{x^2+1}{x^2-1} = \frac{x^2+1}{(x+1)(x-1)}$ has two vertical asymptotes at $x = -1$ and $x = 1$, respectively.

(c) $f(x) = \cot x = \frac{\cos x}{\sin x}$ has infinitely many vertical asymptotes at $x = n\pi \quad \forall n \in \mathbb{Z}$.



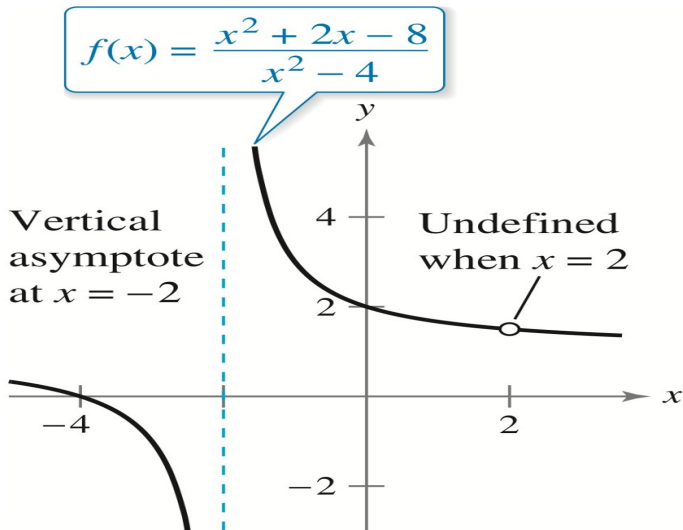
to asymptote. to
Example³: Let $f(x) = \frac{x^2 + 2x - 8}{x^2 - 4}$. Then

$$f(x) = \frac{(x+4)(x-2)}{(x+2)(x-2)} = \frac{x+4}{x+2} \text{ for } x \neq -2 \text{ and } \underline{x \neq 2.}$$

$\Rightarrow f$ has a vertical asymptote at $\underline{x = -2}$.



Example 3 的示意圖 (承上頁)



Thm 1.15 (Properties of Infinite Limits)

Suppose that $\lim_{x \rightarrow c} f(x) = \pm\infty$ and $\lim_{x \rightarrow c} g(x) = L \neq 0$.

- ① $\lim_{x \rightarrow c} [f(x) \pm g(x)] = \pm\infty$.
- ② $\lim_{x \rightarrow c} [f(x)g(x)] = \pm\infty$ if $L > 0$.
- ③ $\lim_{x \rightarrow c} [f(x)g(x)] = \mp\infty$ if $L < 0$.
- ④ $\lim_{x \rightarrow c} \frac{g(x)}{f(x)} = 0$.



Example 5 Find the limits.

$$(a) \lim_{x \rightarrow 0} \left(1 + \frac{1}{x^2}\right) = 1 + \lim_{x \rightarrow 0} \frac{1}{x^2} = \infty \quad \text{because } \lim_{x \rightarrow 0} \frac{1}{x^2} = \infty.$$

$$(b) \lim_{x \rightarrow 1^-} \frac{x^2 + 1}{\cot \pi x} = \frac{2}{-\infty} = 0 \quad \text{because } \lim_{x \rightarrow 1^-} \cot \pi x = -\infty.$$

$$(b) \lim_{x \rightarrow 1^-} \frac{x^2 + 1}{\cot \pi x} = 0 \quad \text{because } \lim_{x \rightarrow 1^-} \cot \pi x = \lim_{x \rightarrow 1^-} \frac{\cos \pi x}{\sin \pi x} = \frac{-1}{+0} = -\infty.$$

$$(c) \lim_{x \rightarrow 0^+} 3 \ln x = 3 \left(\lim_{x \rightarrow 0^+} \ln x \right) = -\infty.$$

$$(d) \lim_{x \rightarrow 0^-} \left(x + \frac{1}{x}\right) = 0 + \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty.$$



